

Legal Query Synthesis from Business Processes for Agentic Retrieval

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Abstract—Ensuring robust cross-domain retrieval in Legal Information Retrieval (LIR) remains challenging due to the substantial semantic gap between informal business process descriptions and highly formalized regulatory language. This work introduces an agentic query synthesis framework that bridges this gap by transforming structured business process descriptions into diversified legal search queries. The proposed method employs multiple specialized large language model (LLM) rewriting agents that generate complementary query perspectives, including legal terminology rewriting, compliance obligations, risk scenarios, and contractual interpretations. These queries are subsequently used in a weighted ranked reciprocal fusion (WRRF) retrieval pipeline combined with lexical and semantic retrieval methods.

To increase coverage further, an iterative LLM-based critic refines queries based on initial retrieval results, allowing context-aware query adaptation. The framework is evaluated on a manually annotated gold standard corpus with three business process use cases such as insurance claims, Know-Your-Customer (KYC) procedures and a GDPR-driven data breach. Experimental results show the query diversification with iterative refinement can significantly improve the recall with competitive precision compared to baseline LIR systems. The results demonstrate the scalability and transparency of agentic query rewriting as an effective approach to improve regulatory retrieval from complex business process

Index Terms—Legal Information Retrieval, Large Language Models, Query Diversification, Query Rewriting, Learning-to-Rank

I. INTRODUCTION

Businesses across all domains must comply with extensive regulatory requirements to avoid reputational damage, significant fines, operational restrictions, and potential legal sanctions. Identifying relevant regulatory obligations, however, remains a complex and resource-intensive task due to the vast volume of legal texts and the inherent difficulty in manually mapping specific business activities to applicable rules. This creates a clear need for advanced Legal Information Retrieval (LIR) systems capable of effectively bridging business process descriptions with compliance-relevant regulatory content to support legal, compliance, and domain experts. A primary challenge in developing such systems lies in the substantial vocabulary mismatch between the two domains. Legal texts employ highly formalized, technical language and standardized phrasing, while business process descriptions typically use operational, domain-specific terminology. An effective LIR system must therefore bridge this semantic gap by extracting implicit compliance needs from business descriptions and making missing contextual information explicit for accurate retrieval. One of the most effective

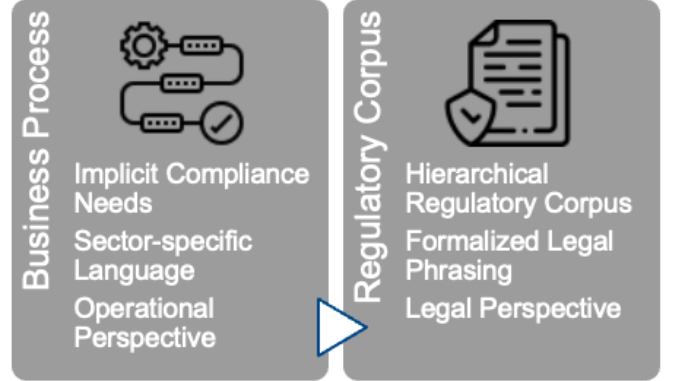


Fig. 1: Domain Shift from Business Process to Regulatory Corpus

techniques employed in state-of-the-art Legal Information Retrieval frameworks is query diversification (also referred to as query enrichment or reformulation). By generating multiple rephrased versions of a query from different legal, compliance-based and risk-based perspectives, this approach makes implicit requirements explicit and ensures broader coverage of relevant regulatory aspects. In this report, a structured approach will be presented that leverages query diversification to automatically synthesize high-quality search queries for an LIR system. Starting from business process, subprocess, and task-level descriptions, the method generates diversified queries that are subsequently used to retrieve regulatory requirements directly relevant to the described activities. This approach will be embedded in an agentic retrieval pipeline and applied to and evaluated by testing it on example business processes including a travel insurance claim process, a "Know-Your-Customer" use case and a highly GDPR-relevant data breach scenario.

II. RELATED WORK

This section reviews existing approaches for constructing LIR systems for large corpora, as well as LIR query diversification and the theoretical basis for query diversification methods.

A. LIR System Design

In both [1] and [2] frameworks for LIR given a user question have been proposed. Both make use of multi-agent systems to generate search queries to cover multiple legal perspectives of the user query which increases recall [2]. Combined with iterative retrieval rounds, they replace the step of traditional retrieval based only on the user query, as it is the case in [3]. The results of this retrieval are then in

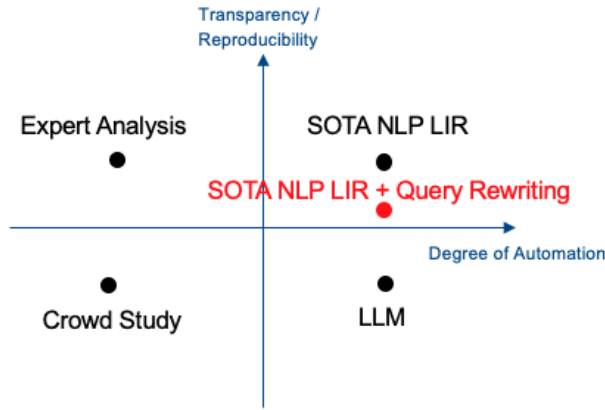


Fig. 2: Comparison of different LIR methods

turn either re-ranked [2] or summarized by another agent [1]. In [3] multiple methods for LIR have been compared, such as Expert Analysis, SOTA-NLP-LIR, an LLM-based approach to classify the document and a crowd-study. The different approaches have been evaluated based on the dimensions Transparency and reproducibility as well as their degree of automation, as shown in 2.

B. Query Diversification

According to [4] query expansion, routing and transformation are among the most promising methods for reliable legal retrieval systems, as well as Summary Augmented Chunking (SAC) and a hierarchical version of which. The advantage of SAC lies in its global context which is included into every chunk so that the semantic gap resulting from implicit information and missing context as described in 1 can be mitigated.

C. Critical Summary

To the best of my knowledge, the existing literature is limited by at least one of the following aspects:

- 1) The specific use case of Legal Information Retrieval for business processes as in [3] has not yet been combined with query diversification methods for the retrieval step.
- 2) Agentic end-to-end pipelines for LIR have not been built specifically to support the practical annotation of business processes.

D. Contributions

To address the previous limitations, the key contributions of this paper are the following:

- This work introduces a framework for LIR for Business Processes that includes agentic query rewriting techniques combined with SOTA retrieval methods based on the SOTA-NLP-LIR approach from [3].
- This report compares different approaches to query diversification and query merging and compares them to the existing baseline as published in [3].

- This work includes a visual business process annotator, that can be used to enrich a business process diagram with relevant information about regulatory requirements from a preselected document corpus. Furthermore a practical end-to-end system design for such an application is proposed.

III. ARCHITECTURE DESIGN DECISIONS

This work proposes a Large Language Model (LLM)-based framework for LIR which is based on the SOTA-NLP-LIR from the paper [3]. In the following I will explain the original implementation and explain the new additions to it. The original implementation of the SOTA-NLP-LIR approach aims to identify the top k most relevant documents from a regulatory requirement corpus for a given business process, subprocess or task-level description. It is comprised of two different configurations:

- 1) (1) Lexical Retrieval (BM25) followed by
(2) Semantic Reranking (Cross Encoder *cross-encoder/ms-marco-MiniLM-L-6-v2*)
- 2) (1) Semantic Retrieval (Bi-Encoder *multi-qa-MiniLM-L6-cos-v1*) followed by
(2) Semantic Reranking (Cross Encoder *cross-encoder/ms-marco-MiniLM-L-6-v2*)

Unlike in the original system, the newly proposed framework does not perform retrieval immediately using an extract from the business process as search query, but first applies a step of **query diversification** and query enrichment with a multi-agent rewriting approach. After obtaining separate retrieval rankings for each query, an **aggregated ranking** is created. To solve the problem of missing context as addressed by SAC [4], a **critic** is used after the retrieval steps to again adapt the search queries given context from the corpus.

A. Query Diversification

The initial query stems from a description in a structured business process representation with different granularities (process, subprocess, task). In order to cover several legal dimension of the original query in the retrieval step, unexpressed compliance implications have to be made explicit. For this step, using an LLM is a logical solution, since they provide reasoning capabilities and context-knowledge that exceeds information contained in original query and hence can be used to enrich it. Based on the original query q_0 , the queries $(q_0, q_1, q_2, q_3, q_4)$ are later used for retrieval. The agent design is aimed at creating a general approach to LIR, that can be applied to a majority of LIR tasks. For this, four separate rewriting agents have been chosen. In more specialized cases, there could also be different combinations of agents, that would improve the query rewriting quality. The agents' distinct tasks will be explained in the following.

- **Legal Terminology Rewriter:** This agent is the most general of the agents. It fulfills the function of bridging the vocabulary mismatch between the original query and the legal terminology used in the corpus. It serves as a translator between the two domains.

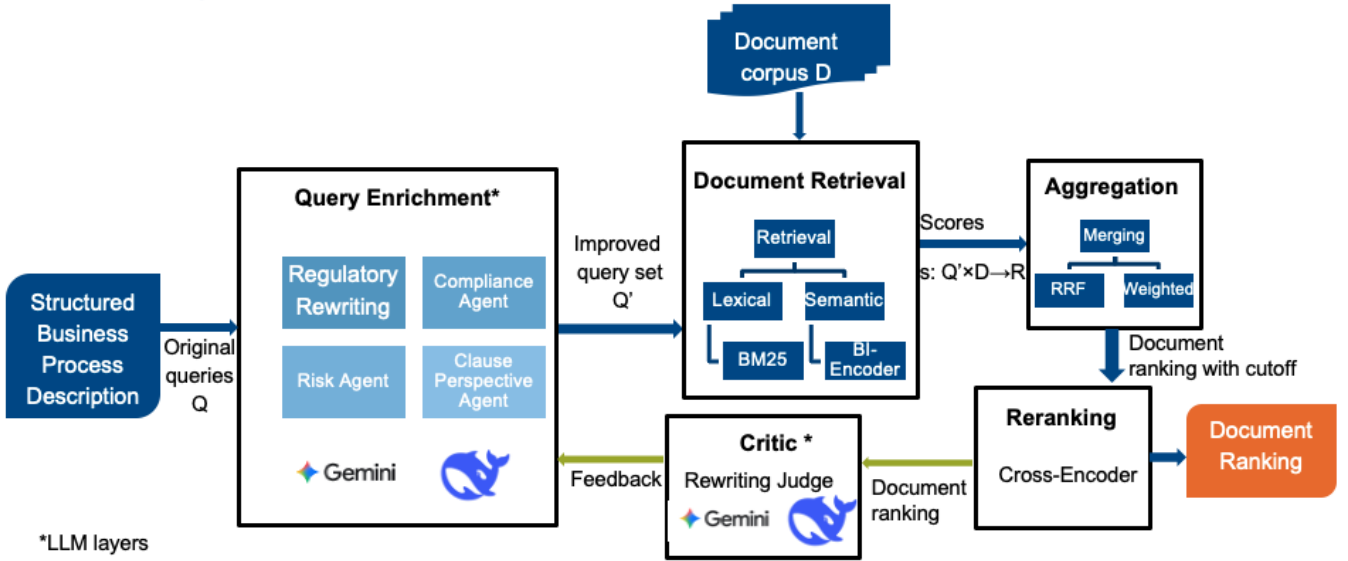


Fig. 3: Overview of the proposed framework based on query rewriting to retrieve regulatory requirements for business processes.

- **Regulatory Compliance Agent:** This agent reframes the original query from a perspective centered around implicit compliance aspects, such as obligations, restrictions and reporting requirements.
- **Scenario Risk Agent:** This agent focuses on finding potential failure modes, edge cases and legal violations that are an implicit possibilities of the original query. It provides a perspective on the legal risks of business operations.
- **Clause/Contract Perspective Agent:** This agent rephrases the original query in terms of implicit contract obligations.

B. Aggregating Query Rankings

For each search query dimension the retrieval is performed, so for one original query that we need to map to relevant regulatory requirements, we obtain 5 retrieval scores $S_i(d)$ for each document in the corpus. Since BM25 output scores are not normalized, they are harder to compare and the need arises to work with the ranking positions $R_i(d)$ of the documents instead. A common approach to merge rankings of documents is Ranked Reciprocal Fusion (RRF), where k is an arbitrarily selected smoothing constant.

$$\text{RRF}(d) = \sum_{i=0}^4 \frac{1}{k + r_i(d)}$$

It is not automatically given however, that all queries are equally important for the final retrieval result, since e.g. the original query is closest for the original intent, while e.g. the Clause/Contract agent might be useful in some cases, but also contributes only to generate noise in some, when the content of the original query does not have any contractual dimension. To express this in the RRF-scoring function, it

can be extended by adding weights w_i for each query.

$$\text{WRRF}(d) = \sum_{i=0}^4 w_i \frac{1}{k + r_i(d)}$$

Given the need for weights, there are two different approaches for selecting them that I want to focus on in this project. The first one being hand-picking weights, and the second using a statistical Learning-to-Rank model to calculate optimal weights given a labeled training dataset.

1) *Handpicking Weights:* This method is based purely on observations and reasoning, as of which query might contain the most relevant information for LIR. It is reasonable to assume that the original query q_0 and the general legal terminology rewriting query q_0 would generate the least noise and are therefore the safest choices to assign higher weights to. The other 3 queries are all about roughly equally informative and risky to generate noise. Hence I arrived at the weight vector $W_{handpicked}$.

$$W_{handpicked}^T = [0.5 \quad 0.29 \quad 0.07 \quad 0.07 \quad 0.07]$$

2) *Linear Model: Learning to Rank:* In order to pick a linear model for this specific task we need to consider, that we are working with a binary labels that either belong to the class of relevant documents for a query or non-relevant ones. Since we are searching weights for a linear model, we assume a linear decision boundary for our features. Hence we can chose a logistic regression for this task. For this we assign the following scores to a query, document pair.

$$x_i(q_i, d) = \begin{cases} \frac{1}{k + r_i(d)} & \text{if document } d \text{ is retrieved via } q_i \\ 0 & \text{otherwise} \end{cases}$$

Then on a labeled dataset, we train a logistic regression.

$$P(y = 1 \mid q, d) = \sigma \left(\beta_0 + \beta^T \mathbf{x}(q, d) \right)$$

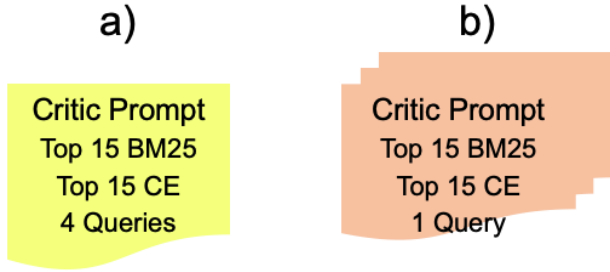


Fig. 4: Prompts for the LLM critic (a) For Gemini 3.1-Pro, with all queries being improved at once (b) For local Deepseek R1, each query being changed individually

Finally the weights are made non-negative and normalized to be usable in the WRRF-function.

$$w_i = \frac{\max(\beta_i, 0)}{\sum_j \max(\beta_j, 0)}$$

The weights that we obtained this way now give us information about which queries are the most accurate for creating a ranking.

C. LLM Critic

After the first retrieval round, a critic agent is being prompted to improve the search queries, given additional context about the corpus and the first retrieval results as well as about all generated queries. This can lead to adaptations to the corpus and improved coverage. Here the first $k=15$ documents from both retrieval stages will be provided in the prompt to give a wide range of context about the corpus.

For small local models used in the evaluation such as Deepseek-R1 with 9 Billion parameters, the model is not consistent enough to provide structured output for the improved queries without fine-tuning, so instead of sending one prompt with all queries, the critic is asked to improve the queries one by one as shown in 4.

D. Model Choices

To test the capabilities of the system under best possible circumstances, Google’s Gemini 3.1 Pro is used as an LLM, since it is a hosted frontier model with over 1 Trillion parameters and reasoning capabilities.

To compare it to a free open weight model that can be run locally, the pipeline is also evaluated with Deepseek R1, which also has reasoning capabilities for comparativeness with Gemini 3.1 pro. The model has 9 Billion parameters and can run locally on a MacBook Pro with M3 and 16GB RAM.

IV. EVALUATION PIPELINE

To measure the performance of the system, it is compared to a Gold Standard dataset (GSD) that has been created in [3] and consists of two business process use cases that have manually been annotated by domain experts. This GSD serves as the ground truth for the evaluation of the results, so it can be compared to the results from [3], which serve

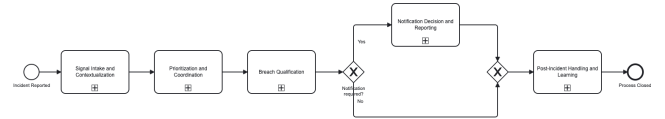


Fig. 5: Data Breach Business Process

TABLE I: Ablation study on Query Diversification (UC 1)

Level	Method	Accuracy	Precision	Recall
Process	Base	0.75	0.19	0.46
Process	<i>Simple RRF</i>	0.77	0.14	0.29
Process	<i>Handpicked Weights</i>	0.76	0.14	0.29
Subprocess	Base	0.95	0.15	0.32
Subprocess	<i>Simple RRF</i>	0.92	0.05	0.15
Subprocess	<i>Handpicked Weights</i>	0.92	0.05	0.15
Task	Base	0.97	0.09	0.23
Task	<i>Simple RRF</i>	0.96	0.05	0.18
Task	<i>Handpicked Weights</i>	0.96	0.05	0.19

as a baseline to compare the query diversification approach to.

For this work, a new use case has been synthetically created and manually annotated to serve as additional data used for evaluation. To be comparable to the other 2 existing ones from [3], also 10% of the corpus is relevant for at least one query. The use case is a data breach scenario consisting of 5 subprocesses and 20 tasks. The corpus consists of the GDPR laws and the EU AI Act. There are 169 documents in the corpus, 17 of which are relevant to at least one task.

Since the usual LIR application has a human in the loop, it is especially important to provide all necessary documents in a pre-retrieved selection, to not miss any compliance requirements. Therefore, the most important metric which is meant to be improved by query diversification is recall.

V. RESULTS

In the evaluation the naive methods of RRF and handpicked weights without an LLM critic perform slightly below the baseline in [3] for all 3 use cases and across all 3 granularities except for the process level in use case 2 using Gemini 1.5 Pro to rewrite the queries.

The reason for this is the noise introduced by the queries, which is not accounted for by setting the weights appropriately or introducing additional context.

To evaluate the linear model, Use Case 1 and 2 have been split in 2 halves randomly. One of which is used as a training dataset and the other as a validation dataset. Use Case 3 is independent and serves as part of the validation dataset to verify generalizability of the calculated weights of the model. The best performing model uses weights calculated across all three levels, as opposed to using level-specific weights.

The results for the trained weights in combination with the LLM critic are significantly better than for the simple RRF approaches. The ablation study has been carried out on both Gemini 1.5 Pro and DeepSeek R1. Both models beat the baseline for Use Cases 2 and 3. However, for Use Case 1, there is a structural difficulty, since a lot of relevant documents are not from a Code of Conduct that is not phrased in formal legal language. For Use Case 1 Gemini

TABLE II: Ablation study on Query Diversification (UC 2)

Level	Method	Accuracy	Precision	Recall
Process	Base	0.74	0.11	0.23
Process	<i>Simple RRF</i>	0.72	0.14	0.45
Process	<i>Handpicked Weights</i>	0.72	0.14	0.45
Subprocess	Base	0.94	0.09	0.23
Subprocess	<i>Simple RRF</i>	0.89	0.05	0.18
Subprocess	<i>Handpicked Weights</i>	0.89	0.06	0.22
Task	Base	0.97	0.09	0.23
Task	<i>Simple RRF</i>	0.94	0.04	0.10
Task	<i>Handpicked Weights</i>	0.94	0.04	0.11

TABLE III: Ablation study on Query Diversification (UC 3)

Level	Method	Accuracy	Precision	Recall
Process	Base*	0.48	0.04	0.24
Process	<i>Simple RRF</i>	0.48	0.04	0.24
Process	<i>Handpicked Weights</i>	0.48	0.04	0.24
Subprocess	Base*	0.79	0.01	0.04
Subprocess	<i>Simple RRF</i>	0.79	0.01	0.02
Subprocess	<i>Handpicked Weights</i>	0.79	0.01	0.02
Task	Base*	0.89	0.01	0.04
Task	<i>Simple RRF</i>	0.89	0.00	0.01
Task	<i>Handpicked Weights</i>	0.89	0.00	0.01

1.5 Pro only performs better than the baseline on the lowest level.

Gemini 1.5 Pro’s performance beats DeepSeek R1’s, which is to be expected, given that Gemini 1.5 Pro is a state-of-the-art model, while DeepSeek R1 is a smaller local model with 9 billion parameters.

The results show that query diversification in combination with iterative-retrieval rounds can not only improve recall significantly, but also precision and accuracy.

VI. LIMITATIONS

The limitations of this project have been mainly 3 different ones: the availability of data, the goal of generalizability and the time available for the implementation.

- 1) **Data:** In order to achieve good statistical results for the weight optimization it is required to use enough data so that the result can be evaluated correctly. Currently the same use cases 1 and 2 are used for training and validation and only use case 3 can serve as a proof for generalizability.
- 2) **General Approach:** Due to some Use Cases having particular structural characteristics or needs for different legal perspectives, better results could be achieved by creating different rewriting agents depending on the particular use case and corpus. Especially in Use Case 1 the performance varies significantly from the other 2 Use Cases.
- 3) **Time:** Due to the local DeepSeek R1 LLM being slow when running on an Apple Silicon M3 chip, the feedback cycle for it is very long, leading to me not being able to include it into the ablation study for all methods but only to the most promising one (LLM critic + Weights from logistic regression). Furthermore the prompts for the rewriting agents could also be tested with non-reasoning models and a CoT approach or a few-shot approach in the future.

TABLE IV: Ablation study on LLM Critic + Optimal Weights (UC 1)

Level	Method	Accuracy	Precision	Recall
Process	Base	0.75	0.19	0.46
Process	<i>Gemini 1.5 Pro</i>	0.761	0.160	0.327
Process	<i>DeepSeek R1</i>	0.769	0.180	0.367
Subprocess	Base	0.95	0.15	0.32
Subprocess	<i>Gemini 1.5 Pro</i>	0.929	0.083	0.263
Subprocess	<i>DeepSeek R1</i>	0.927	0.067	0.211
Task	Base	0.97	0.09	0.23
Task	<i>Gemini 1.5 Pro</i>	0.965	0.084	0.260
Task	<i>DeepSeek R1</i>	0.964	0.071	0.219

TABLE V: Ablation study on LLM Critic + Optimal Weights (UC 2)

Level	Method	Accuracy	Precision	Recall
Process	Base	0.74	0.11	0.23
Process	<i>Gemini 1.5 Pro</i>	0.752	0.270	0.871
Process	<i>DeepSeek R1</i>	0.740	0.250	0.806
Subprocess	Base	0.94	0.09	0.23
Subprocess	<i>Gemini 1.5 Pro</i>	0.907	0.183	0.550
Subprocess	<i>DeepSeek R1</i>	0.900	0.150	0.450
Task	Base	0.97	0.09	0.23
Task	<i>Gemini 1.5 Pro</i>	0.948	0.180	0.415
Task	<i>DeepSeek R1</i>	0.944	0.133	0.308

VII. END-TO-END PIPELINE

In this section, a hypothetical End-to-End pipeline for this system will be proposed. Currently a webapp to annotate business processes with their relevant regulatory requirements exists, which is built with FastAPI to serve the backend and React to render the frontend. If such a system would be built to be used in production by legal departments, many aspects would need to be changed. First of all there should be a selection from multiple large legal corpora that can either be managed by an organization or scraped from the web. Furthermore the hosting of such a system would require the deployment of the services in the cloud to ensure scalability and maximum up-time, for example as a Container in the Google Cloud Kubernetes Engine. Furthermore there would be a need to export the retrieval results from within the app and edit them collaboratively.

VIII. CONCLUSION AND OUTLOOK

The results have shown that the retrieval quality benefits from query diversification and having multiple re-writers. However, compared to pure LLM-based retrieval the results remain worse; however latency is higher and the costs are lower while the transparency of the results is higher as well.

There are more methods to be tried out however, and the rewriting agent configuration can be changed entirely based on the given use case.

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TABLE VI: Ablation study on LLM Critic + Optimal Weights (UC 3)

Level	Method	Accuracy	Precision	Recall
Process	<i>Base*</i>	0.48	0.04	0.24
Process	<i>Gemini 1.5 Pro</i>	0.506	0.170	1.000
Process	<i>DeepSeek R1</i>	0.506	0.170	1.000
Subprocess	<i>Base*</i>	0.79	0.01	0.04
Subprocess	<i>Gemini 1.5 Pro</i>	0.842	0.213	0.681
Subprocess	<i>DeepSeek R1</i>	0.832	0.187	0.596
Task	<i>Base*</i>	0.89	0.01	0.04
Task	<i>Gemini 1.5 Pro</i>	0.912	0.147	0.530
Task	<i>DeepSeek R1</i>	0.909	0.130	0.470

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